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Typical Issues in Medical Imaging *(and Deep Learning)*

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Disclaimers

- This is an **overview** of typical issues and potential solutions.
- I work mostly with brain MRI data, and I will illustrate with examples I encounter in my projects and the community.
- I am **not** a Deep Learning expert.
- I will likely not know the answers in great depth (but someone in the room probably does!).

What is the number 1 headache in dealing with medical images in your projects?



<https://app.wooclap.com/PUTSJHJ?from=instruction-slide>

Can we detect and prognosticate COVID-19 from chest imaging?

A systematic review of machine-learning models for diagnosis and prognosis of COVID-19 from chest imaging screened the entire literature.

2,212 papers screened

415 passed first screen

62 fully reviewed

0 clinically
usable

0

models of potential clinical use — all had methodological flaws or high risk of bias.

Roberts, M., Driggs, D., Thorpe, M. *et al.* Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nat Mach Intell* **3**, 199–217 (2021).

Main reasons

1

Biased, poor-quality data

Many used pediatric scans as the “non-COVID” class and adult scans as COVID — so the model learned to tell children from adults, not detect disease.

2

Poor ML methodology

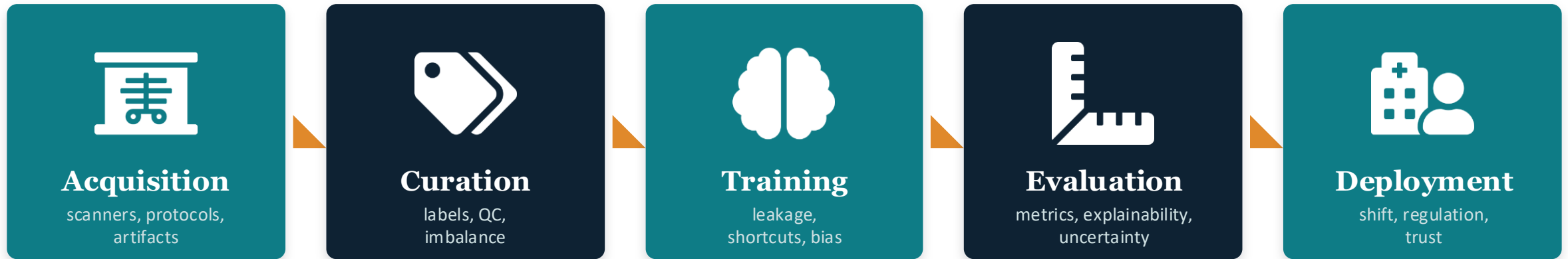
Training and testing on the same data, improper splits and data leakage — with no external validation on independent data.

3

Poor reproducibility & reporting

Source data, code and documentation often unavailable; high risk of bias; little clinician involvement in study design.

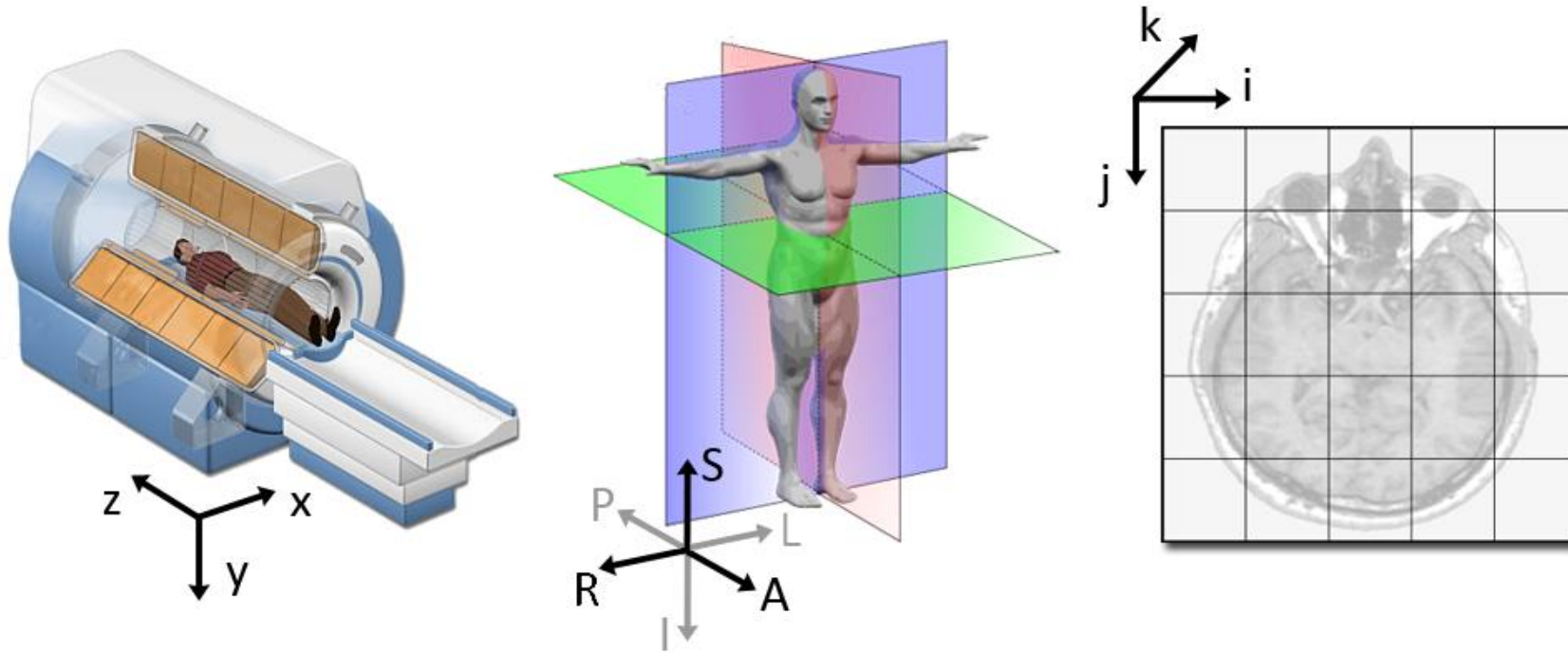
A typical machine learning pipeline





Data

Technical headaches



$$\begin{pmatrix} y_x \\ y_y \\ y_z \\ 1 \end{pmatrix} = \begin{pmatrix} R_x \Delta_{Col} & C_x \Delta_{Row} & 0 & T_x \\ R_y \Delta_{Col} & C_y \Delta_{Row} & 0 & T_y \\ R_z \Delta_{Col} & C_z \Delta_{Row} & 0 & T_z \\ 0 & 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} x_i \\ x_j \\ 0 \\ 1 \end{pmatrix}$$

$\mathbf{y} = \mathbf{Mx}$

Technical headaches

1

GEOMETRY

Shape & space

3D/4D volumes don't fit in memory; anisotropic voxels, varying field-of-view and orientation conventions must be aligned first.

2

INTENSITY & FORMAT

Pixels & headers

MRI intensities are arbitrary and scanner-dependent; wide bit-depth needs windowing; DICOM / NIfTI / slide metadata is inconsistent.

3

SCALE & COMPUTE

Size & I/O

Gigapixel slides and TB-scale cohorts make data loading the bottleneck; tiny targets in huge volumes cause voxel-level imbalance.

4

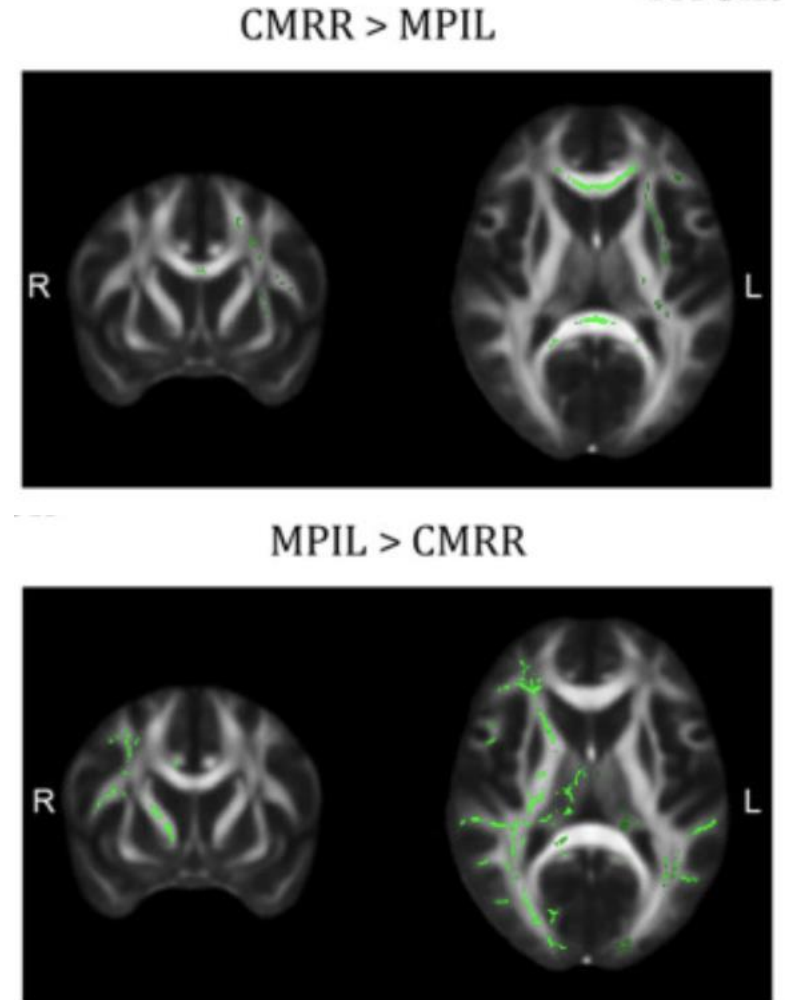
PREPROCESSING & QUALITY

Clean-up & artifacts

Many pipeline-dependent steps, missing sequences and artifacts — each a source of variation, bugs, and even shortcuts.

Same brain, different look?

- **121** healthy participants underwent **two** matched diffusion MRI protocols in **two** different scanners.
- dMRI features differed up to 14%.
- These biases vary between brain regions.
- Scanner differences **may exceed and considerably confound physiological** effects such as ageing.



Distribution shifts and poor generalization

1

ACQUISITION

Scanner & sequence

Vendor, model, field strength (1.5T vs 3T), coils, reconstruction, sequence parameters and slice thickness — or stain and digitizer in pathology — all reshape appearance.

2

SITE / PROTOCOL

Pipelines

Each center adds its own protocols and preprocessing, so “site” bundles many small shifts at once — which is why inter-site gaps usually exceed scanner-alone effects.

3

POPULATION

Who is scanned

Age, sex, race / ethnicity, body habitus and case-mix differ across cohorts, shifting the distribution of how anatomy and pathology actually present.

Poor generalization – Some solutions

1

INPUT LEVEL

Harmonization & preprocessing

Remove scanner/site nuisance variation before it ever reaches the model.

e.g. ComBat-style harmonization · intensity & stain normalization · image-to-image style transfer · standardized pipelines

2

REPRESENTATION LEVEL

Foundation models + fine-tuning

Learn features robust enough that site & scanner matter less, then adapt.

e.g. Large-scale self-supervised pretraining · task / domain fine-tuning · linear probing

3

TRAINING LEVEL

Domain generalization & adaptation

Make training itself span — and discount — domains.

e.g. Diverse multi-site data · augmentation simulating scanner variation · domain-invariant / adversarial features

4

DEPLOYMENT LEVEL

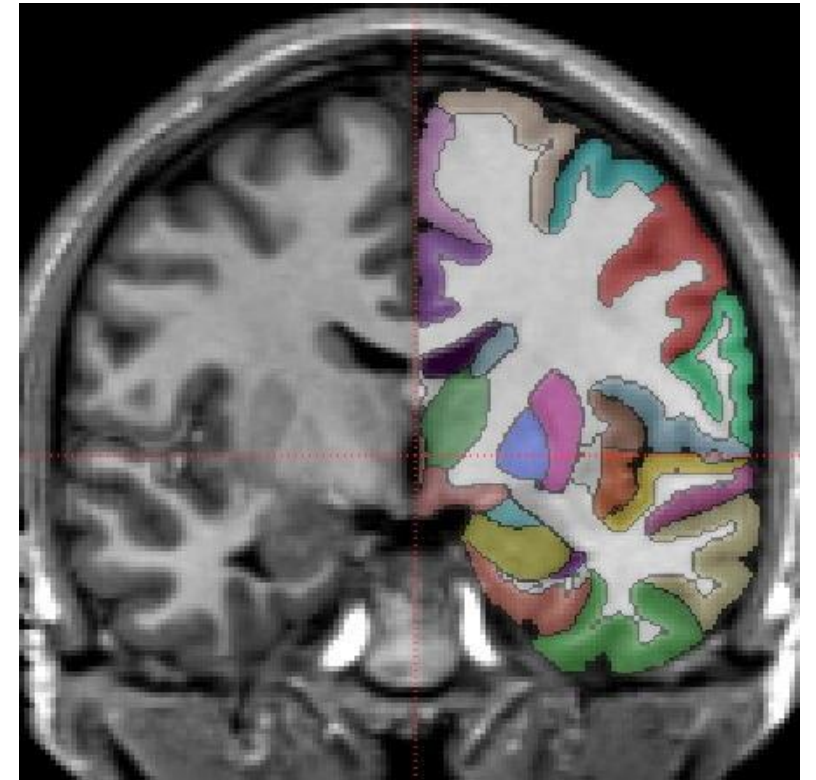
Test-time adaptation & monitoring

Cope with shift after the model ships — and notice when it happens.

e.g. Test-time normalization · OOD detection & uncertainty · post-deployment drift monitoring

Back to brain MRI – Data scarcity

- What is the largest dataset of manual subcortical segmentation from MRI?
- Commercial Neuromorphometrics (2015)
 - **114** scans of subjects ranging in age from 5 to 96
 - The cost to manually label one scan is \$2449
- Open science (2023)
 - **100** scans newer data, newer protocol
 - Very biased sample (young adults - 22–35)



Main causes of data scarcity

1

ANNOTATION COST

Labels need scarce experts

Radiologists and pathologists are costly and time-limited — and many tasks need dense pixel- or voxel-level labels, not one tag.

Slow, costly, doesn't scale.

2

PRIVACY & REGULATION

Data stays siloed

HIPAA / GDPR, IRB approval and consent keep images inside institutions. De-identifying scans and reports is itself a burden.

Hard to share even when it exists.

3

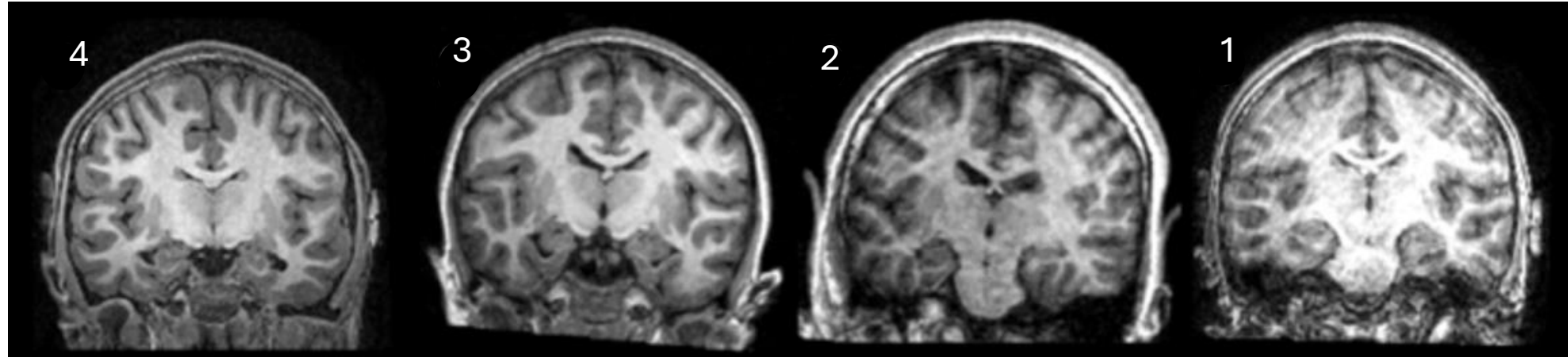
RARITY OF THE TARGET

Few positive cases

Many pathologies of interest are genuinely uncommon, so disease examples are scarce.

Class imbalance at the source.

Example class imbalance – Auto QC



QC Score	1/2	3	4
# cases	9	90	125

- Good for your study, not so good for training your classifier

Scarcity and imbalance – Some solutions

1

LEARN FROM UNLABELED DATA

Foundation models, SSL & transfer

Pretrain on large unlabeled image pools, then fine-tune with only a few labels.

e.g. self-supervised pretraining · transfer learning · domain fine-tuning

→ **answers: annotation cost**

2

MAKE RARE CASES COUNT

Data augmentation & synthesis

Expand and rebalance training data, including synthetic examples of rare findings.

e.g. geometric / intensity augmentation · GAN & diffusion synthesis

→ **answers: rarity (+ privacy via synthetic data)**

3

TRAIN WITHOUT MOVING DATA

Federated learning

Train across sites sharing model weights, not images; pair with privacy safeguards.

e.g. federated training · differential privacy · secure aggregation

→ **answers: privacy & regulation**

4

SPEND LABELS WISELY

Weak supervision & active learning

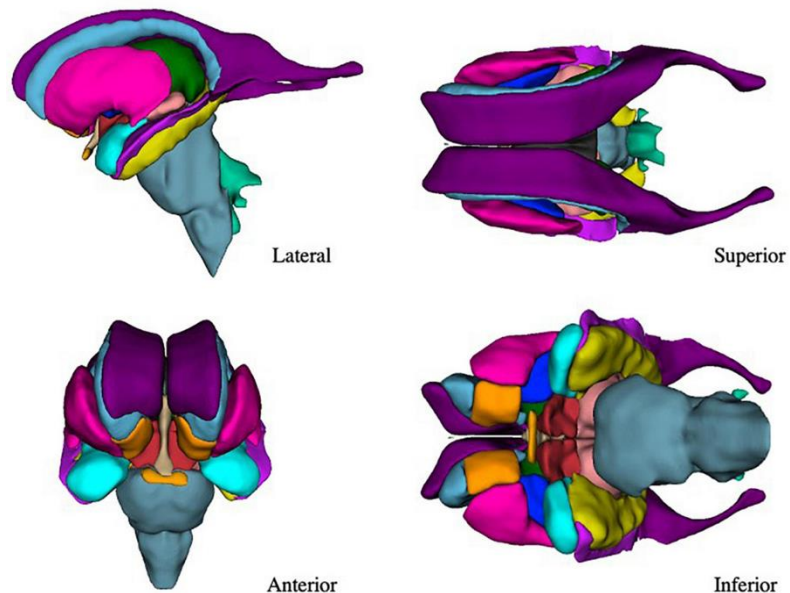
Get more from cheap or partial labels and prioritize what experts annotate.

e.g. report / NLP-mined labels · noisy labels · active & semi-supervised learning

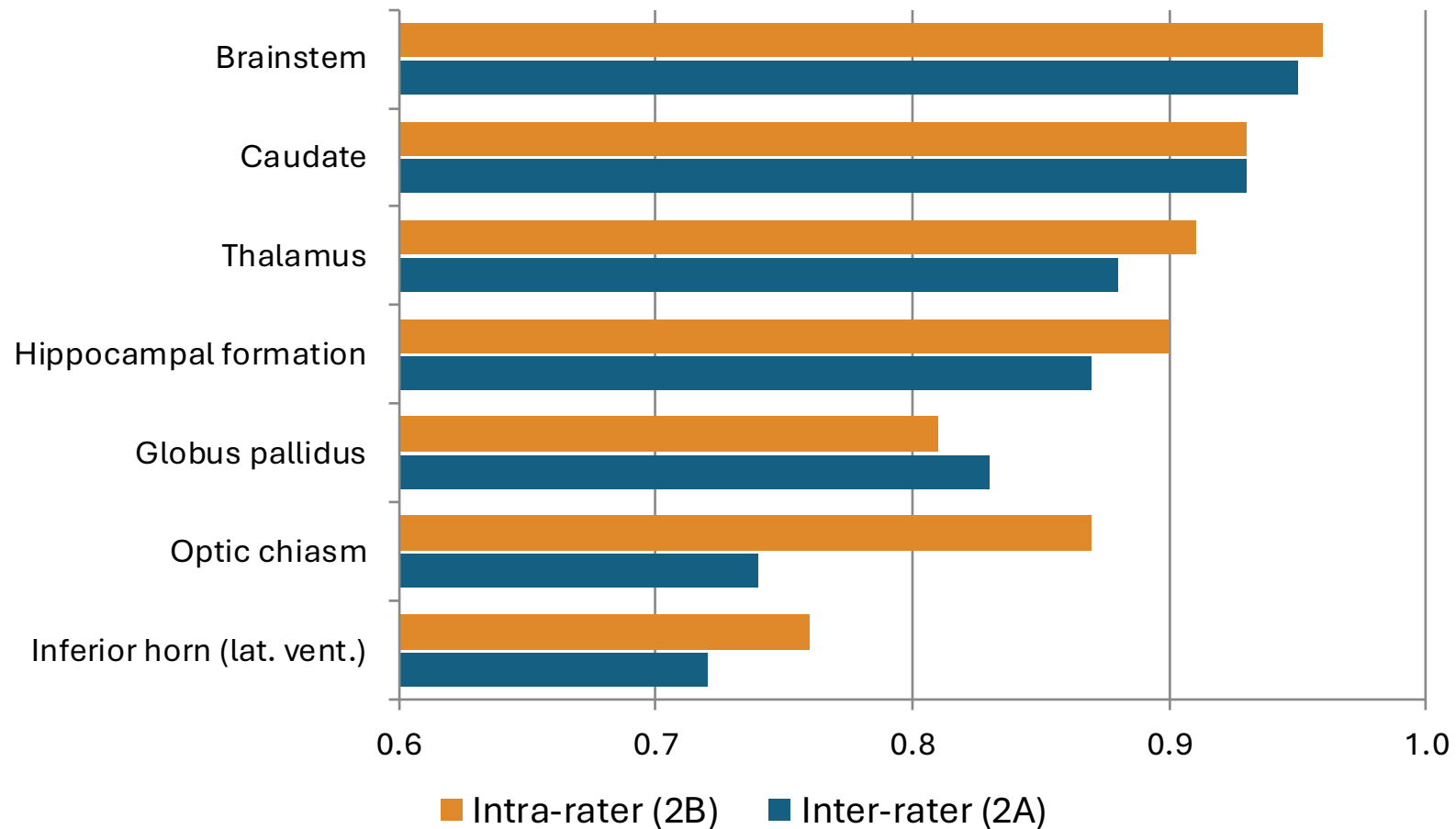
→ **answers: annotation cost**

Brain MRI – Silver standard

- We often train with the output of another algorithm.
- That convenient “silver standard” creates a ceiling in performance, and propagates its biases.



Brain MRI – “Gold” standard



Reliability is hard!

Dice overlap ranges between **0.72** and **0.96**, even when the same person does the same task twice



Does ground truth exist?

1

Inter-rater variability

Two experts segment the same lesion differently. Their mutual Dice is your realistic upper bound — often well below 1.0.

2

Weak labels from reports

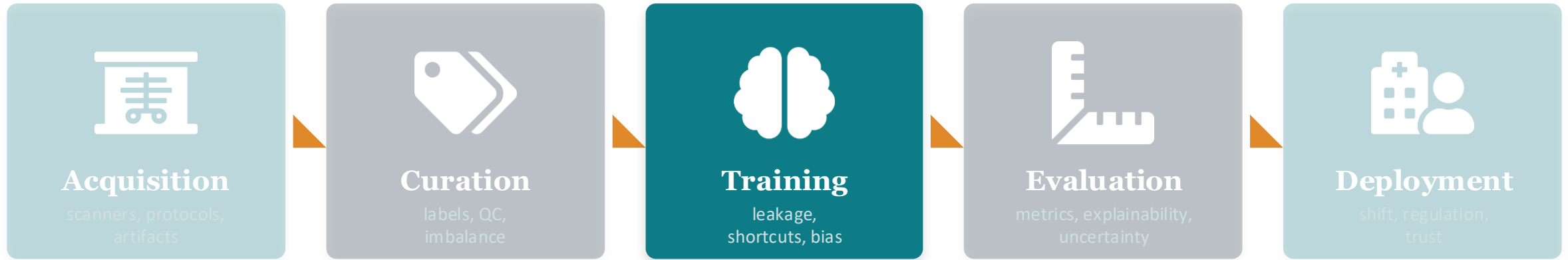
CheXpert and others derive labels from report text — propagating ambiguity and NLP error into training.

3

Fuzzy boundaries

Where exactly does a tumor end? Sub-voxel disagreement becomes systematic label noise in segmentation.

Mitigate: multiple annotators with reported agreement (κ , pairwise Dice) · consensus / STAPLE fusion · noise-robust losses · a small expert-adjudicated test set · treat report-mined labels as weak supervision.



Training

Data leakage, training shortcuts, biased models

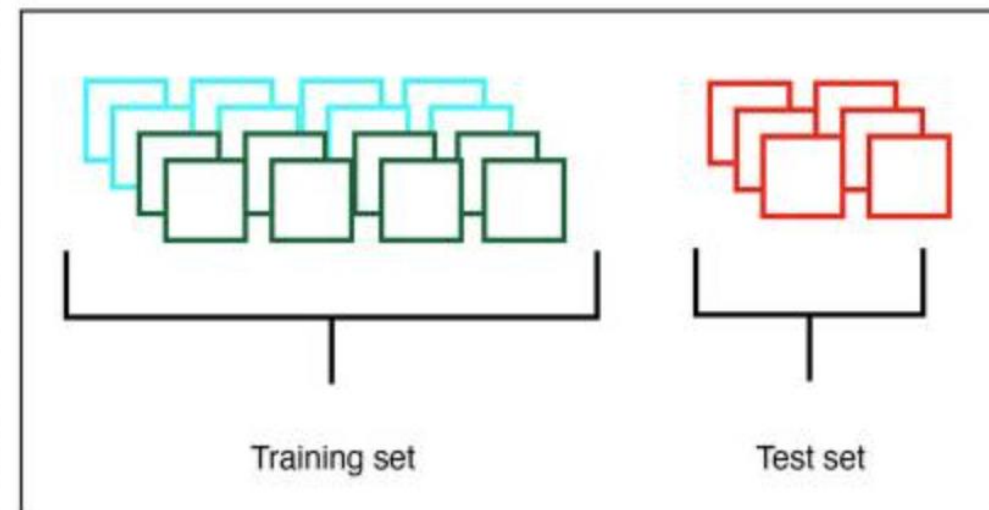
Brain MRI

- **Classification task** (healthy vs disease) in public Alzheimer and Parkinson sMRI datasets from MRI.
- 3D data splitting into 2D slice-level to train 2D CNN.
- Slice-level CV erroneously boosted the average slice level accuracy on the **test set** by 29% to 55% compared to subject-level splits.
- **Inflation** from true values (50–70%) to near-perfect scores (95–100%).

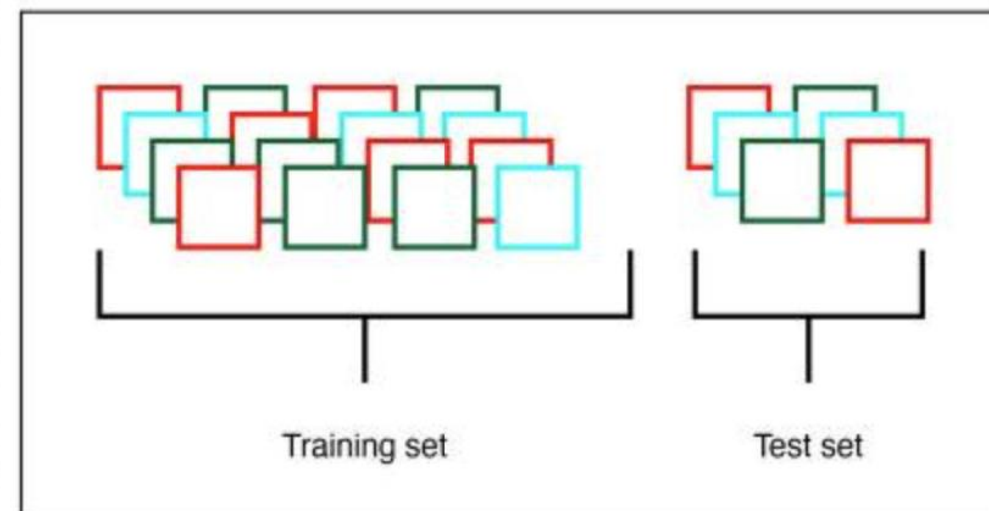
Yagis, E., *et al.* Effect of data leakage in brain MRI classification using 2D convolutional neural networks. *Sci Rep* **11**, 22544 (2021).

CNN model training and validation (nested CV)

(a) Subject-level split



(b) Slice-level split



Data leakage

1 Patient-level

One patient's many images — slices, time points, both eyes, repeat scans — split across train and test.

2 Preprocessing

A step that learns from data (normalization, feature selection, harmonization) is fit on the full set before splitting.

3 Target / label

The model keys on an artifact correlated with the label instead of the pathology itself.

4 Temporal

Data from after the prediction point leaks in — or time is split randomly instead of train-past / test-future.

5 Site / batch

Site correlates with the label, so random splits let the model recognize site signatures. External-site validation is the real test.

6 Duplicates

The same or near-identical image lands in both splits — overlapping public sources or augmented copies crossing the boundary.

Be careful with family structure (sibling, twins, etc.)

Data leakage – Some safeguards

Split first, before any data-dependent step, and touch the locked test set once

1 Patient-level

Split by patient, not image — keep all slices, timepoints and repeat scans in one fold. Verify zero overlap.

2 Preprocessing

Fit normalization, feature selection and harmonization on the train fold only; wrap it in a pipeline and tune with nested CV.

3 Target / label

Strip label proxies before training — burned-in text, device markers, post-outcome fields. Distrust any 'too-predictive' feature.

Data leakage – Some safeguards (cont.)

Split first, before any data-dependent step, and touch the locked test set once

4 Temporal

Split chronologically — train on the past, test on the future. Fix the prediction time point; never shuffle across time.

5 Site / batch

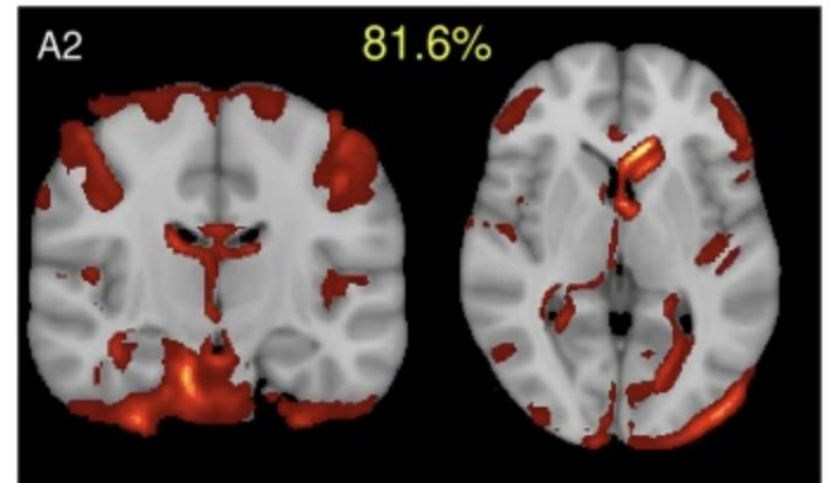
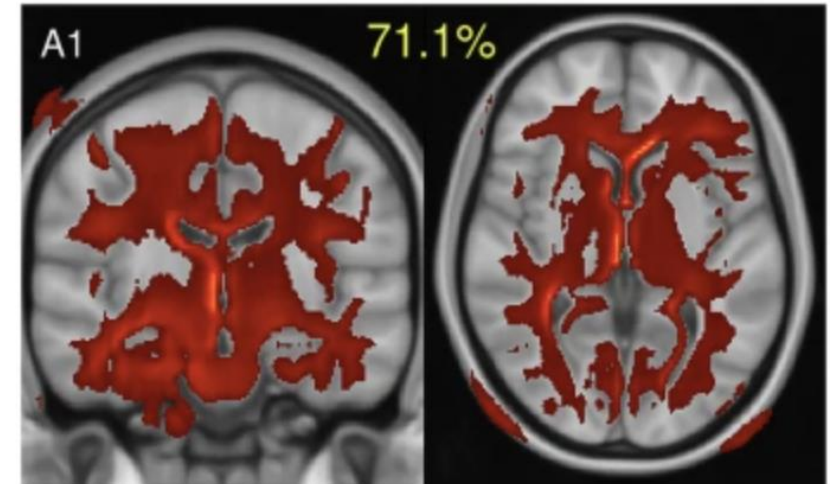
Hold entire sites out (leave-one-site-out / external validation). Fit any harmonization on training sites only.

6 Duplicates

De-duplicate the whole dataset before splitting; keep augmented copies on one side; be cautious with family structure (e.g., twins)

Brain MRI – What is driving my classifier?

- Classification task: Alzheimer vs. Healthy
- 3DCNN trained with or without skull stripping
- AUC is improved by **10%** but is the performance of the models driven by brain structures or by image preprocessing?



Attribution Maps

Tinauer C, et al. Skull-stripping induces shortcut learning in MRI-based Alzheimer's disease classification. Insights Imaging. 2025 Dec 22;16(1):283.

Shortcut learning

The model reaches high accuracy on a cue that correlates with the label but isn't the pathology. **Common sources of that cue:**

1 Scanner / acquisition

Vendor, field strength, reconstruction and sequence leave a signature the model reads instead of the disease.

2 Site / institution

One center supplies most positives, or formats images its own way — so the model learns the hospital.

3 Text & markers

Burned-in text, dates, view labels and L/R laterality markers ride along with the image.

4 Support devices

Chest tubes, lines, pacemakers and clips co-occur with disease — an effect of diagnosis, not its cause.

5 Preprocessing artifacts

Skull-stripping, masking, padding, field-of-view and intensity conventions become the cue.

6 Demographics & composition

The model infers age, sex or race — or positives and negatives come from different populations.

Shortcut learning – Some solutions

1

INPUT

Remove the cue

Crop and mask burned-in text, markers and background; standardize preprocessing.

e.g. mask markers · standardized pipelines

2

TRAINING DATA

Make the cue unreliable

Train on diverse, multi-site data and balance the sources, with augmentation, so no shortcut consistently predicts the label.

e.g. multi-site data · augmentation · harmonization

3

EVALUATION

Test where it breaks

Validate on external sites; stress-test by hiding the suspected cue and watching if the score drops.

e.g. external validation · per-subgroup metrics · occlusion test

4

INSPECTION

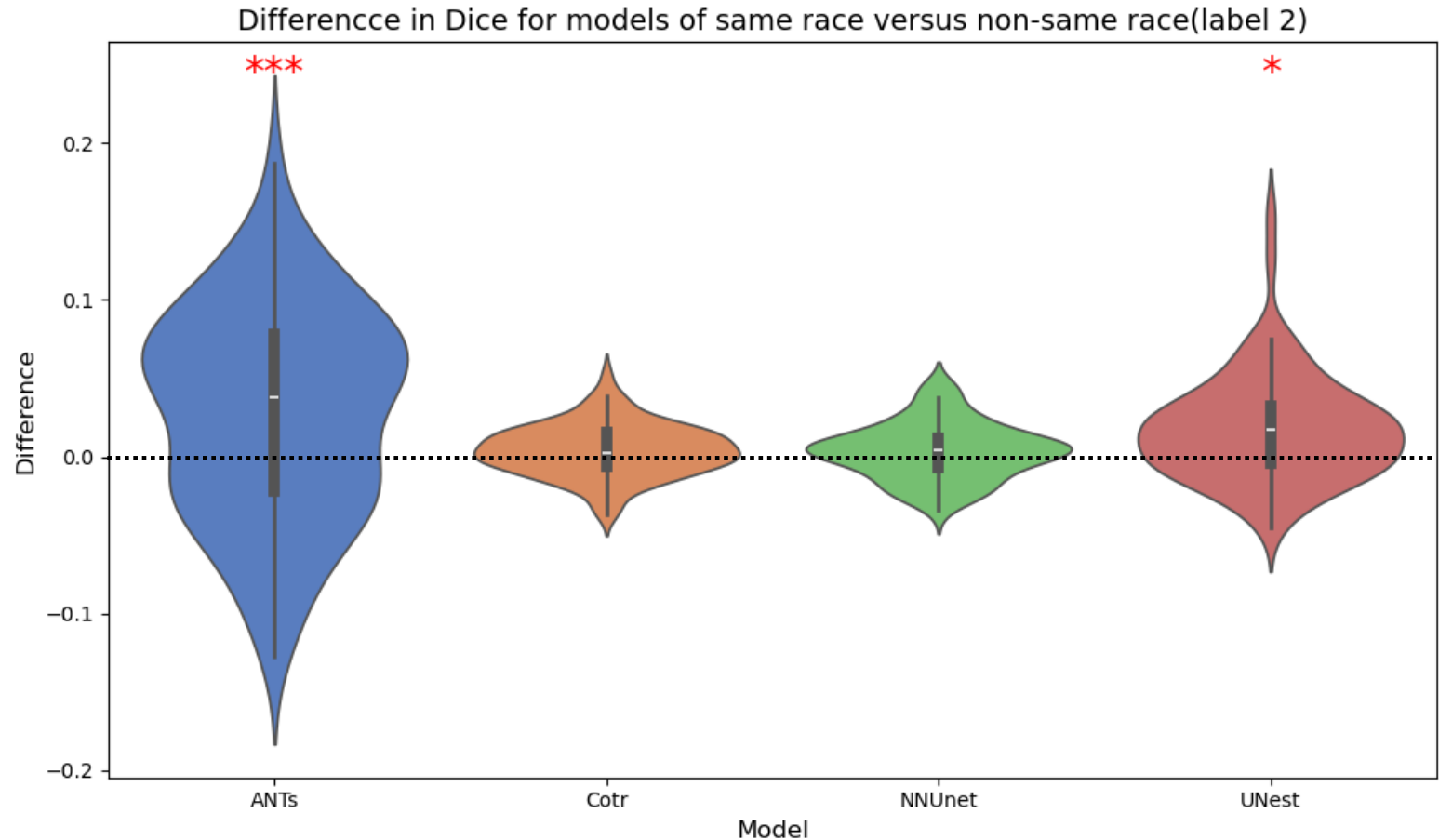
See what it uses

Check that the model attends to anatomy, not artifacts, with saliency maps and simple probes.

e.g. saliency maps · “can it predict the site?” probe

Brain segmentation – Bias

- Brain **segmentation** task
- 4 groups: White/Black Male/Female.
- Trained with one group only.
- Tested the difference in performance depending on whether the training and testing groups were of the same race.



Bias and Fairness – It's complicated

Bias is where inequity comes from; fairness is how we judge the outcome. But “fair” has many definitions — **and equalizing is not always the right clinical goal.**

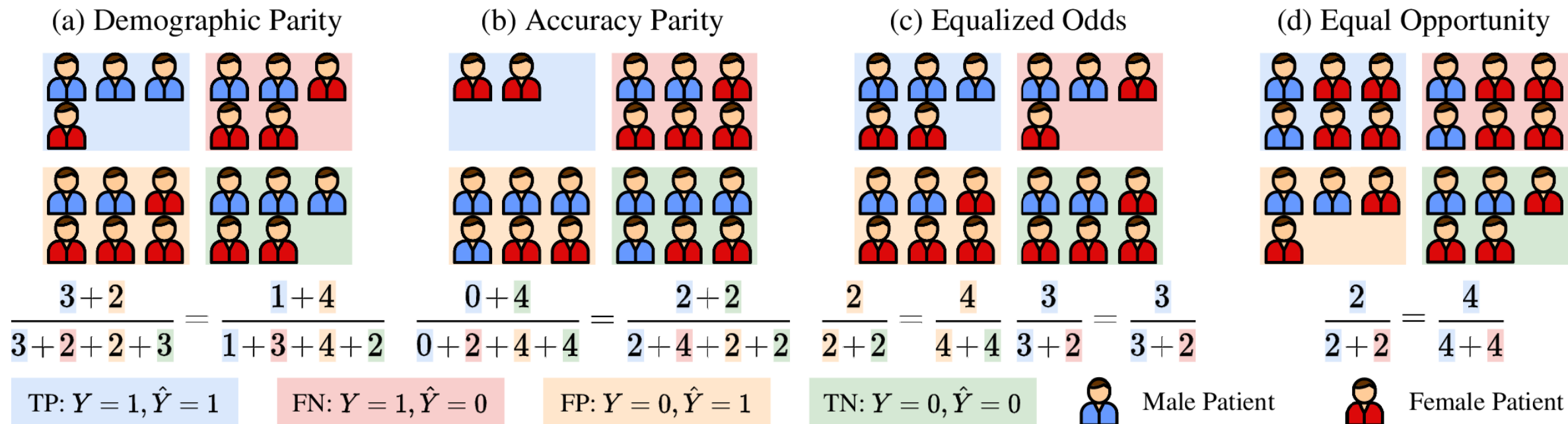
Many definitions that can conflict

- ▶ Demographic parity, equalized odds, equal opportunity, calibration — each is a different rule.
- ▶ They are mathematically incompatible: you generally cannot satisfy them all at once.
- ▶ “Make it fair” is underspecified until you choose which definition fits the clinical question.

Sometimes fairness isn't the goal

- ▶ Equalizing can “level down” — worsening the better-served group instead of lifting the other.
- ▶ Real differences exist (prevalence, anatomy by sex/age); forcing parity can cause under-diagnosis.
- ▶ The aim is the best decision per patient — a protected attribute is sometimes clinically valid.

Measuring Fairness



Metric	Formula	Explanation
Demographic Parity (DP)	$P(\hat{Y}=1 \mid A=0) = P(\hat{Y}=1 \mid A=1)$	The model outcome should not be affected by any sensitive attribute.
Accuracy Parity (AP)	$P(\hat{Y}=Y \mid A=0) = P(\hat{Y}=Y \mid A=1)$	The model should have equal accuracy among subgroups.
Equalized Odds (EqOdds)	$P(\hat{Y}=1 \mid A=0, Y=y) = P(\hat{Y}=1 \mid A=1, Y=y), y \in \{0,1\}$	The model should have equal TPR and FPR among subgroups.
Equal Opportunity (EqOpp)	$P(\hat{Y}=1 \mid A=0, Y=1) = P(\hat{Y}=1 \mid A=1, Y=1)$	The model should have equal TPR among subgroups.

Fairness – Causes and (partial) remedies

Stage	Where unfairness comes from	What you can do
Data	Under-represented groups	Add or rebalance training data (<i>augmentation</i>)
	Real anatomical differences (e.g., sex)	Generate synthetic examples (<i>causal synthesis</i>)
	Inconsistent labels (<i>annotation noise</i>)	Use several annotators
Model	Models chosen on accuracy alone (<i>ERM</i>)	Pick for accuracy and fairness together (<i>multi-objective / DTO</i>)
	Model keys on sex or age, not disease (<i>spurious correlation</i>)	Remove the misleading cue (<i>confounder removal</i>)
Deployment	Bias carried over from pretraining (<i>fine-tuning</i>)	Trim the biased parts of the model (<i>pruning</i>)
	Deployment site looks different (<i>domain gap</i>)	Re-tune to the new site (<i>domain adaptation</i>)

Fairness of AI in Medical Imaging

Medical AI fairness resources

🌟 Upcoming Events

FAIMI Home

Resources

Newsletter

MELBA Special Issue

📅 Past Events

<https://faimi-workshop.github.io/>

What's the common thread behind the failure modes we just covered?

Scarcity, imbalance, shortcut, leakage, ...

Know your data!

IF YOU KNOW IT

Interrogate it

- ▶ Provenance — patient ID, site, scanner, dates
- ▶ How labels were made — raters, reports, consensus
- ▶ How cases were selected — inclusion & referral
- ▶ Class & subgroup balance
- ▶ Duplicates & near-duplicates

Lets you split correctly, stratify for bias, and audit for shortcuts.

IF YOU DON'T

Don't assume it is flawless

- ▶ Assume leakage, shortcuts & bias until disproven
- ▶ Treat public-benchmark scores as possibly contaminated
- ▶ Own a private external / prospective test set
- ▶ Probe — perturbations, ablations, subgroup analysis
- ▶ Don't rely on a single aggregate number

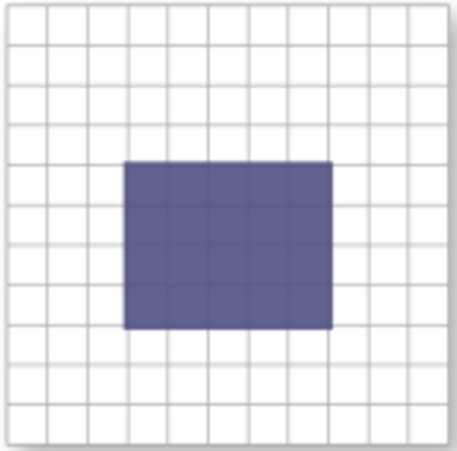
Caution is the substitute for the knowledge you don't have.



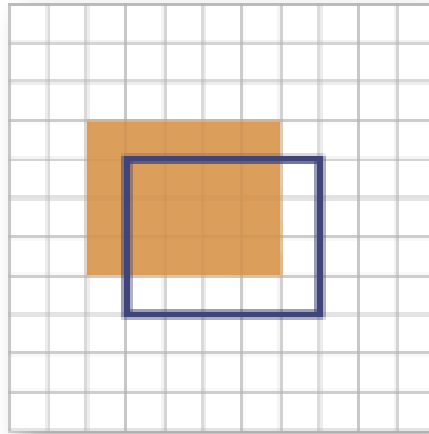
Evaluation & Deployment

Dice = 0.6

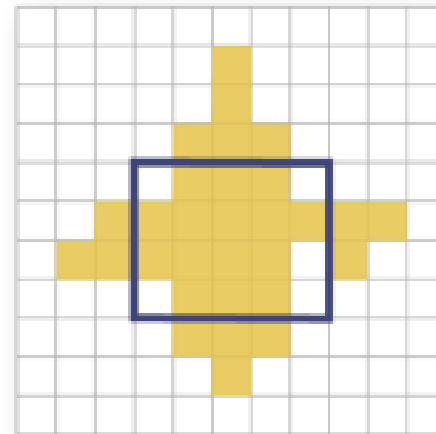
Reference



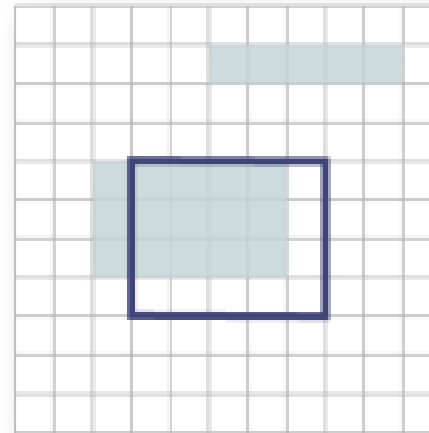
Prediction 1



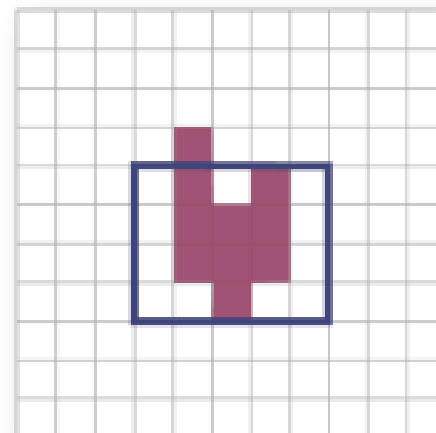
Prediction 3



Prediction 2



Prediction 4



1) BIOPSY IN ALL PATIENTS:

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 30	FN 0
	Negative	FP 75	TN 0

Accuracy = 0.29

BUT

75 unnecessary biopsies (FP)

2) MARKER-BASED DECISION ON BIOPSY

		PREDICTED	
		Positive	Negative
ACTUAL	Positive	TP 20	FN 15
	Negative	FP 60	TN 10

Accuracy = 0.29

BUT

60 unnecessary biopsies (FP)

Cost-benefit analysis:

~9 unnecessary biopsies for
one detected lesion are
acceptable.



**NB metrics relate the benefits of TPs with the cost of FPs by incorporating an exchange rate
(here: 1/9 based on benefit-cost analysis)**

NB = 0.21

>

NB = 0.13

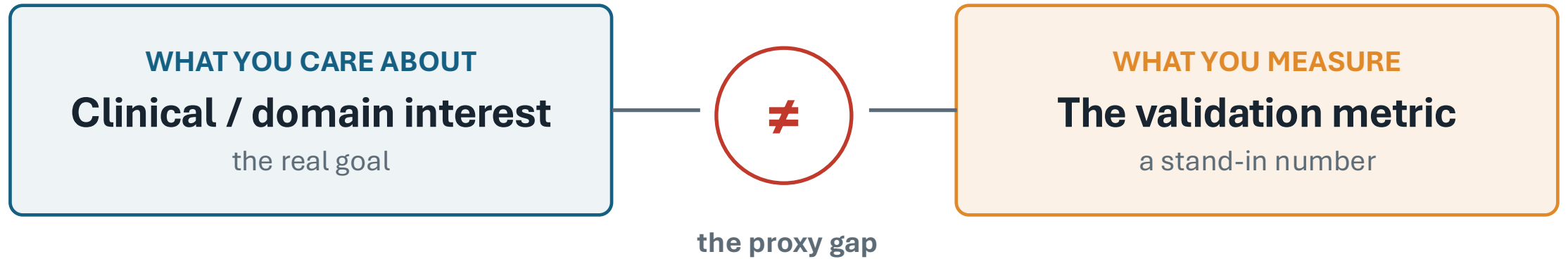
Same Accuracy, but
better clinical utility

Same Accuracy, but
poorer clinical utility

$$\text{Net Benefit} = (\text{TP} - \text{FP} \times w) \div n$$

Performance metrics

A validation metric is only a **proxy** for what you actually care about. Choose it by convention, and the numbers in papers stop reflecting real-world usefulness — a key reason imaging AI fails to translate.



97 distinct metrics found across ~15 years of biomedical-imaging challenges

...each with its own blind spots — and metric choice is rarely matched to the problem.

Main pitfalls in choosing metrics

P1

Wrong problem category

Framing the task wrong — e.g. scoring object detection as segmentation — makes the standard metrics meaningless.

→ **frame the task right first**

P2

Wrong metric

A metric blind to the goal or with poor properties: accuracy under class imbalance, overlap blind to small structures.

→ **match metric to the goal**

P3

Wrong application

Right metric, misused: improper aggregation (per-image vs pooled), ignoring imbalance or uncertainty.

→ **apply & aggregate carefully**

a

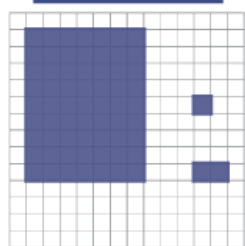
Various pitfalls related to choice of validation metric



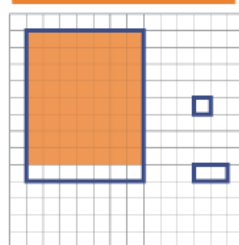
Inappropriate choice of the problem category

Example – object detection confused with semantic segmentation: DSC is strongly biased toward single objects and is therefore not appropriate for measuring the detection of multiple objects

Reference



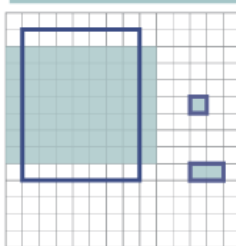
Prediction 1



1 object detected ❌ 3 objects detected ✅

DSC = 0.92 >> DSC = 0.79

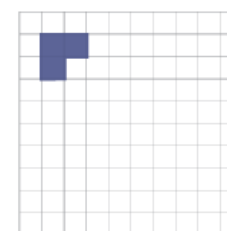
Prediction 2



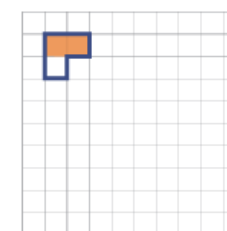
Poor metric selection

Example – neglecting the small size of structures: single-pixel differences can hugely impact the metric scores, which is especially relevant given high inter-rater variability and the non-deterministic nature of AI algorithms

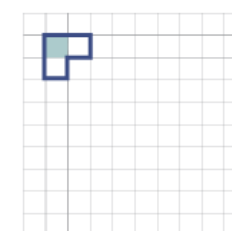
Reference



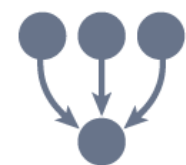
Prediction 1



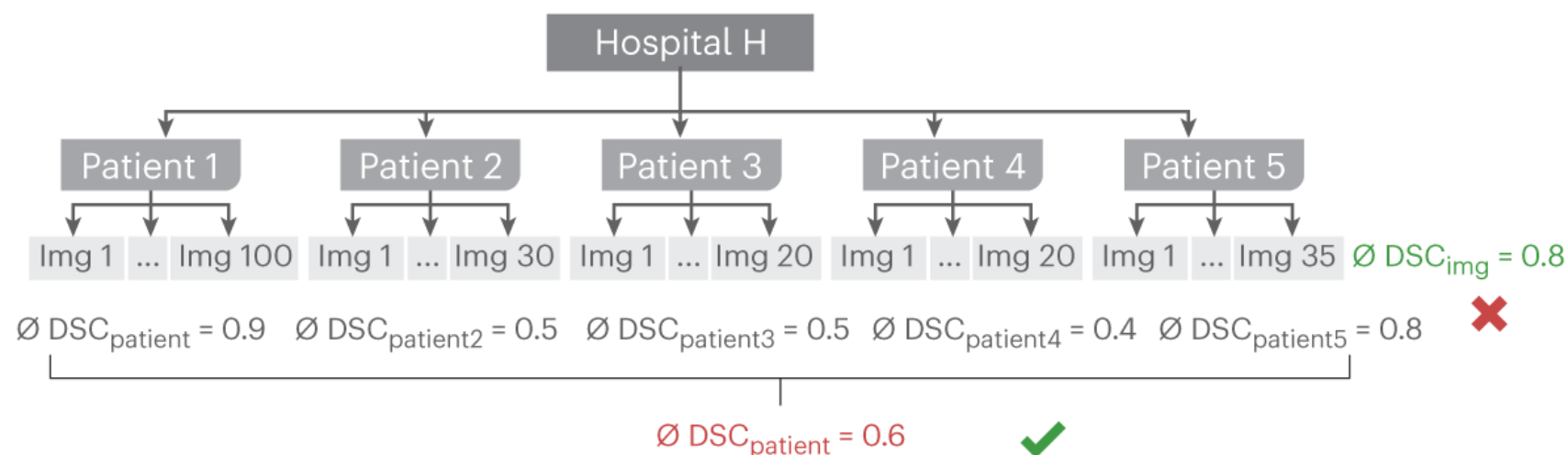
Prediction 2



DSC = 0.80 >> DSC = 0.50



Poor metric application



Example – inappropriate aggregation scheme: hierarchical data structure is often neglected when aggregating metric values, which is especially important for different numbers of cases per variable

Metrics Reloaded

A framework for trustworthy
image analysis validation

(Quick access to Metrics Reloaded main paper



<https://metrics-reloaded.dkfz.de/>

Trust from clinicians and patients

What are the challenges in convincing clinicians and patients that DL models are safe?

Trust from clinicians and patients

1

Interpretability & explainability

Black-box predictions are hard for clinicians to trust or act on — and saliency maps are often unreliable.

2

Uncertainty quantification

Models rarely flag when they are out of distribution or simply unsure — so errors arrive unannounced.

3

Privacy

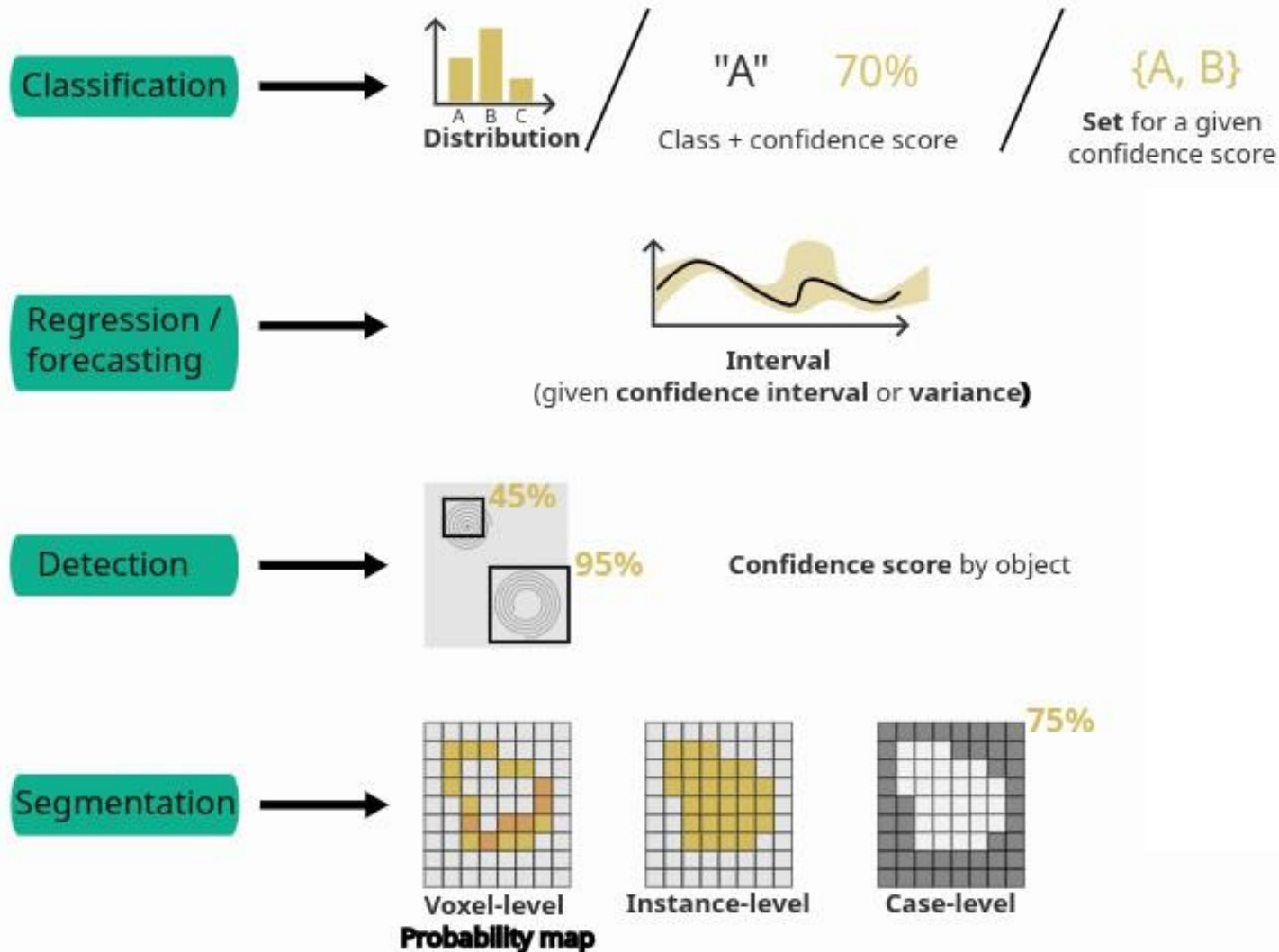
Constraints on sharing data limit training and validation — driving interest in federated learning and harmonization.

Deep Learning for Medical Imaging

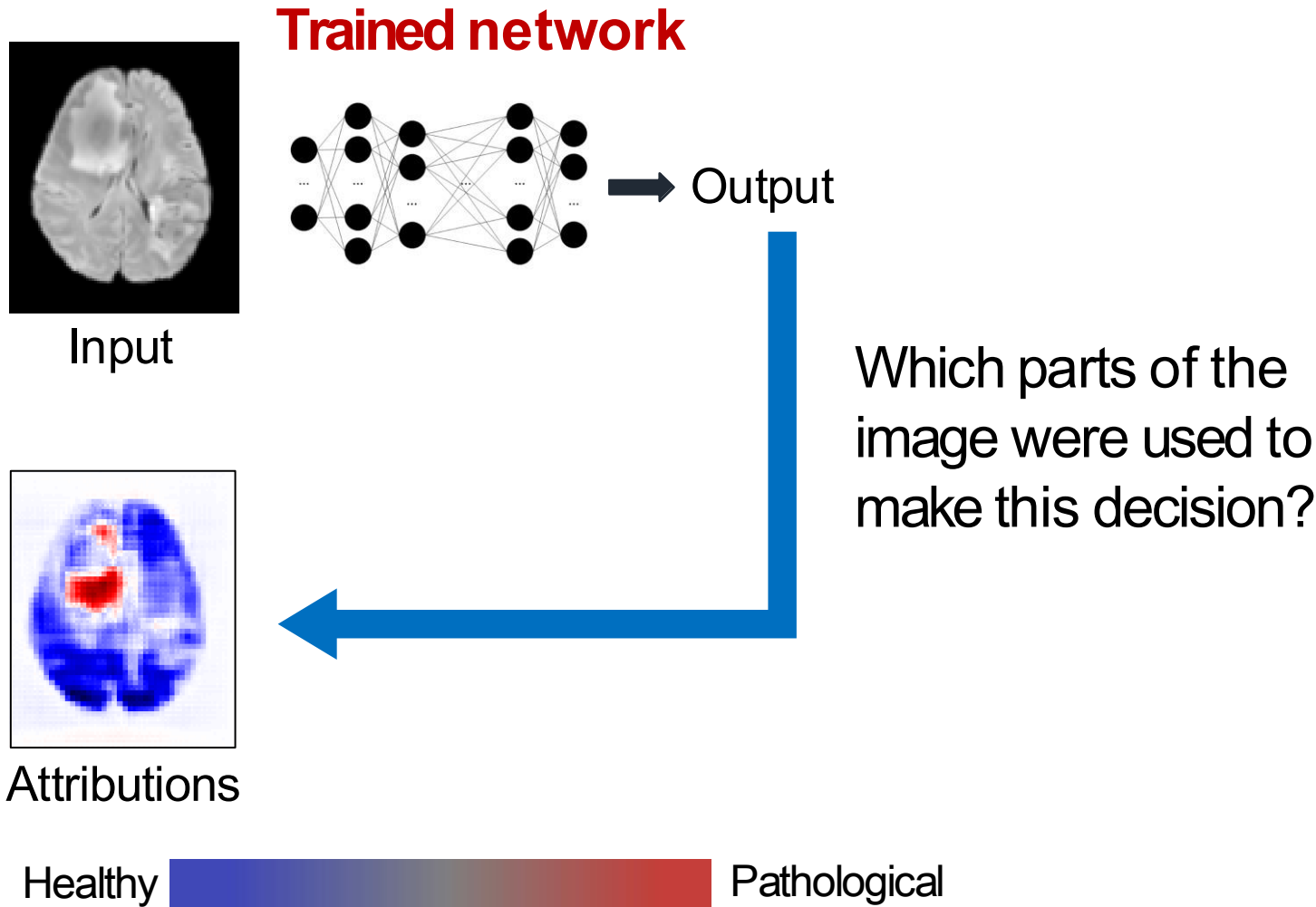
Uncertainties and Explainability

Valentine Wagnier-Dauchelle
CREATIS INSA Lyon

Different task = different uncertainties



Attributions maps



Based on:

- Gradients
- Activations
- Perturbations



Privacy Protection in Medical Imaging

Héber H. Arcolezi (ÉTS Montréal)



Outline



Why privacy
matters

Part I - DATA



Privacy
risks in data



Data-level
protections

Part II - MODELS



Privacy risks
in models



Model-level
protections



Takeaways

Trust from scientists and regulatory entities?

What are the challenges in convincing scientists the methods are sound?

Can we detect and prognosticate COVID-19 from chest imaging?

A systematic review of machine-learning models for diagnosis and prognosis of COVID-19 from chest imaging screened the entire literature.

2,212 papers screened

415 passed first screen

62 fully reviewed

0 clinically
usable

0

models of potential clinical use — all had methodological flaws or high risk of bias.

Roberts, M., Driggs, D., Thorpe, M. *et al.* Common pitfalls and recommendations for using machine learning to detect and prognosticate for COVID-19 using chest radiographs and CT scans. *Nat Mach Intell* **3**, 199–217 (2021).

Reproducibility and clear reporting – the CLAIM checklist

A 44-item standard for transparent, reproducible reporting of medical-imaging AI.



STUDY DESIGN

Design & intended use

Prospective or retrospective, the study goal and hypotheses, and the model's intended clinical role.

CLAIM items 1–6



DATA

Where the data came from

Sources, inclusion / exclusion, preprocessing, de-identification, and how missing data were handled.

CLAIM items 7–13



REFERENCE STANDARD

How ground truth was set

Definition and rationale of the reference standard, who annotated, and inter- / intra-rater variability.

CLAIM items 14–18



PARTITIONS & TEST

Splits done right

How cases were assigned to train / validation / test, kept disjoint, and the intended sample size.

CLAIM items 19–21

CLAIM checklist (continued)

From the model itself through to open reporting — the second half of the checklist:



MODEL

What was built

Architecture described in detail, the software and libraries used, and how model parameters were initialized.

CLAIM items 22–24



TRAINING

How it was trained

The training approach, how the final model was selected, and any ensembling of models.

CLAIM items 25–27



EVALUATION

How it was tested

Metrics with uncertainty, robustness and explainability, and testing on both internal and external data.

CLAIM items 28–34



RESULTS & TRANSPARENCY

Reported openly

Patient flow and subgroup demographics, failure analysis and limitations, plus protocol, code, model and funding.

CLAIM items 35–44

The CLAIM checklist

Checklist for Artificial Intelligence in Medical Imaging (CLAIM): 2024 Update		
Section/Topic	No.	Item
TITLE/ABSTRACT		
	1	Identification as a study of AI methodology, specifying the category of technology used (eg, deep learning)
ABSTRACT		
	2	Summary of study design, methods, results, and conclusions
INTRODUCTION		
	3	Scientific and/or clinical background, including the intended use and role of the AI approach
	4	Study aims, objectives, and hypotheses
METHODS		
Study Design	5	Prospective or retrospective study
	6	Study goal
Data	7	Data sources
	8	Inclusion and exclusion criteria
	9	Data preprocessing
	10	Selection of data subsets
	11	De-identification methods
	12	How missing data were handled
	13	Image acquisition protocol
	14	Definition of method(s) used to obtain reference standard
Reference Standard	15	Rationale for choosing the reference standard
	16	Source of reference standard annotations
	17	Annotation of test set
	18	Measures of inter- and intrarater variability of features described by the annotators
Data Partitions	19	How data were assigned to partitions
	20	Level at which partitions are disjoint

Testing Data Model	21	Intended sample size
	22	Detailed description of model
	23	Software libraries, frameworks, and packages
	24	Initialization of model parameters
Training	25	Details of training approach
	26	Method of selecting the final model
	27	Ensembling techniques
Evaluation	28	Metrics of model performance
	29	Statistical measures of significance and uncertainty
	30	Robustness or sensitivity analysis
	31	Methods for explainability or interpretability
	32	Evaluation on internal data
	33	Testing on external data
	34	Clinical trial registration
RESULTS		
Data	35	Numbers of patients or examinations included and excluded
	36	Demographic and clinical characteristics of cases in each partition
Model Performance	37	Performance metrics and measures of statistical uncertainty
	38	Estimates of diagnostic performance and their precision
	39	Failure analysis of incorrectly classified cases
DISCUSSION		
	40	Study limitations
	41	Implications for practice, including intended use and/or clinical role
OTHER INFORMATION		
	42	Provide a reference to the full study protocol or to additional technical details
	43	Statement about the availability of software, trained model, and/or data
	44	Sources of funding and other support; role of funders

Indicate page and/or line number for each checklist item that is present.

From the best model to the clinic

Regulatory and operational challenges

1

Regulatory approval

FDA, CE/MDR, Health Canada —
including the challenge of
approving models that keep
updating over time.

2

Workflow integration

Fitting into PACS, reporting systems
and clinician time budgets is
nontrivial.

3

Model drift & monitoring

Performance can silently degrade
post-deployment as protocols and
populations change.

Take-home message

Simple pitfalls in principle — often unaddressed

- ▶ Data leakage
- ▶ Shortcut learning
- ▶ Misleading metrics
- ▶ Lack of clinician involvement
- ▶ Weak reporting (CLAIM)

Research opportunities

- ▶ Data scarcity
- ▶ Distribution shift
- ▶ Uncertainty quantification
- ▶ Explainability
- ▶ Privacy
- ▶ Ground truth & fairness